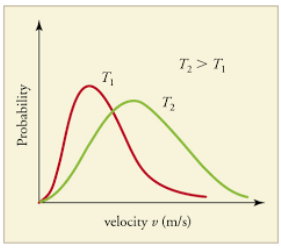
MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth Semester
Discipline Specific Core Course (DSC) - MG4DSCCHE201 - PHYSICAL CHEMISTRY - 1
(2024 ADMISSION ONWARDS)
Duration: 1 hour 30 minutes **Maximum Marks: 50**

Part A - Multiple Choice Questions
(Answer all questions, each carries 1 mark)

1. **1 poise equals**
   Correct Option: (c) 0.1 kg m^{-1} s^{-1}.Poise is a CGS unit of viscosity and in SI units 1 P is equal to 0.1 kg m^{-1} s^{-1}.
2. **The degree of ionization of a weak electrolyte increases with**
   Correct Option: (a) Increase in dilution (c) increase in temperature. As dilution increases, the concentration decreases, leading to greater ionization as per Ostwald's dilution law. Degree of ionization generally increases with temperature if the ionization process is endothermic.
3. **The molal elevation constant (Kb) is also known as**
   Correct Option: (d). Ebullioscopic constant. Kb represents the elevation in boiling point and is therefore called the ebullioscopic constant.
4. **Which is the correct expression for critical pressure in terms of van der Waals constants a and b?**
   Correct Option: (d) $8a/27b^2$. From van der Waals equation, the critical pressure is derived as $Pc = 8a/27b^2$.
5. **Buffer action is based on the principle of**
   Correct Option: (b) Common ion effect. he presence of a common ion suppresses ionization, helping the buffer resist changes in pH.
6. **Henry's law states that at constant temperature, the solubility of a gas in a liquid is**
   Correct Option: (a) Directly proportional to pressure of the gas. According to Henry's law, solubility increases proportionally with the partial pressure of the gas.
7. **What is the SI unit of van der Waals constant 'a'?**
   Correct Option: (a) N m^4 mol^{-2}. The constant 'a' accounts for intermolecular forces and its SI unit is N m^4 mol^{-2}.
8. **According to Ostwald's dilution law for a weak electrolyte, the degree of dissociation is _____ to the square root of concentration.**
   Correct Option: (b).Inversely proportional. The degree of dissociation varies inversely with the square root of concentration for weak electrolytes.
9. **Which among the following is a colligative property?**
   Correct Option: (c). Osmotic pressure. Colligative properties depend only on the number of solute particles, and osmotic pressure is one such property.
10. **At Boyle temperature the second Virial coefficient is**
    Correct Option: (d). Zero. At Boyle temperature, intermolecular forces cancel out, making the second virial coefficient zero.

Part B - Short Answer Questions (Answer any 4, each carries 5 marks)

11. Sketch the Maxwell-Boltzman distribution curve for two different temperatures and explain the influence of temperature on distributionMaxwell-Boltzmann distribution curve and explain temperature effect.

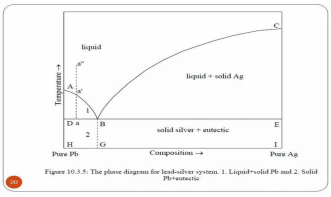


The Maxwell–Boltzmann distribution curve represents the distribution of molecular velocities in a gas, where the x-axis denotes velocity and the y-axis denotes the fraction of molecules. At a given temperature, most molecules have moderate velocities, forming a peak. When temperature increases, the curve becomes broader and flatter, the peak height decreases, and it shifts toward higher velocities, indicating that more molecules possess higher kinetic energy.The value of most probable velocity increases and the fraction of molecules having most propable velocity decreases. Thus, increase in temperature increases average kinetic energy and spreads the velocity distribution. ***((Diagram –2 marks, Any three points with suitable explanation – 3marks)***

12. Define ionic product of water. Derive its expression and state its significance.

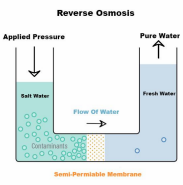
The ionic product of water (Kw) is defined as the product of the molar concentrations of hydrogen ions and hydroxyl ions in water. Water undergoes self-ionization as $H_2O \rightleftharpoons H^+ + OH^-$, and applying the law of mass action gives $Kw = [H^+][OH^-]$. At 25°C, $Kw = 1 \times 10^{-1}$ The significance of Kw lies in its use for calculating pH, distinguishing acidic, basic, and neutral solutions, and forming the basis of acid–base equilibria in aqueous systems. ***(Definition – 2 marks, Derivation – 2 marks , Significance – 1 mark)***

13. Explain in detail one application of a simple eutectic system in metallurgy.



A eutectic system consists of two components that solidify simultaneously at a fixed composition and minimum temperature known as the eutectic point. The phase diagram shows two liquidus curves meeting at the eutectic point. In metallurgy, eutectic systems are used for the separation and purification of metals, such as in the lead–silver system where silver is separated from lead by controlled cooling. ***(Any one correct Application with diagram – 5 marks)***

14. Explain the principle of reverse osmosis. How is reverse osmosis used in the purification of seawater? Mention its advantages



.Reverse osmosis is a process in which external pressure is applied to a solution to force the solvent to move from a concentrated solution to a dilute solution through a semipermeable membrane, opposite to natural osmosis. The setup consists of a membrane that allows only solvent molecules to pass while retaining dissolved salts .This principle is widely used in seawater desalination, where high pressure is applied to seawater to obtain pure drinking water by removing dissolved salts and impurities. ***(Principle – 1.5 marks . Diagram with explanation– 2.5 marks, advantages – 1 mark)***

15. **Determine the pH of a solution obtained by mixing equal volumes of 0.020 N NH_4OH and.020 N NH_4NO_3 (K_b of NH_4OH is 1.80×10^{-5}).**

The mixture of NH_4OH and NH_4NO_3 forms a basic buffer solution consisting of a weak base and its conjugate acid salt. The pH is calculated using the relation

pOH = pKb + log([salt]/[base]).

pKb =−log(1.80×10^{-5}) ≈ 4.74. Assuming equal concentrations of salt and base, pOH = 4.74+ log(0.02 N]/0.02N)= 4.74+log 1= 4.74

pH = 14 − 4.74 = 9.26. ***((*Eqaution 2 Marks, *Substitution and Calculation – 3 marks)***

16. **Evaluate the possibility of separating a mixture of 95.6% ethanol and 4.4% water using simple fractional distillation at constant atmospheric pressure..**

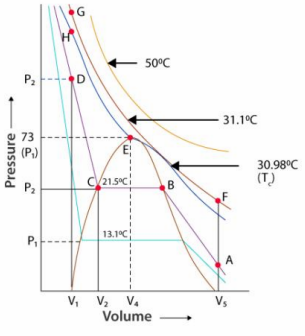
Out of syllabus. **Give full Credit to all.**

Part C - Short Essay Questions (Answer any 2, each carries 10 marks)

17. **Explain Andrew's experiments on isotherms of a real gas and discuss their significance.**

Andrew's studied the behavior of **carbon dioxide (CO_2)** by plotting pressure–volume curves at constant temperatures (isotherms), particularly at **13.1°C, 21.5°C and 31.1°C**, to understand liquefaction of real gases. A fixed amount of CO_2 was enclosed in a strong tube with a piston, and pressure was applied while maintaining constant temperature, recording the corresponding volume changes.

At **13.1°C**, which is well below the critical temperature (31.1°C) of carbon dioxide, Andrew's isotherm shows a clear and detailed pattern of liquefaction with three distinct regions. Initially, in the **gaseous region (A–B)**, CO_2 behaves nearly like an ideal gas, where an increase in pressure leads to a smooth decrease in volume, indicating that the molecules are far apart and intermolecular forces are negligible. As compression continues, the curve reaches a **horizontal portion (B–C)**, which represents the **liquefaction region**. In this region, the pressure remains constant while the volume decreases significantly, showing that gas is being converted into liquid. At point B, liquefaction begins, and between B and C, liquid and vapour coexist in equilibrium. At point C, the entire gas is converted into liquid. This constant pressure is the vapour pressure of CO_2 at 13.1°C. Beyond this, in the **liquid region (C–D)**, further increase in pressure causes only a very small decrease in volume, producing a steep curve because liquids are almost incompressible. Thus, the isotherm at 13.1°C clearly demonstrates the process of liquefaction of a real gas, showing distinct gaseous, transition, and liquid regions, which is characteristic of temperatures below the critical temperature.



At **21.5°C (still below critical temperature but closer to Tc)**, similar behavior is observed, but the **length of the horizontal region decreases**, indicating that liquefaction occurs over a smaller volume range. This shows that as temperature increases, liquefaction becomes more difficult.

At **31.1°C (critical temperature of CO)**, the horizontal portion disappears and is replaced by a **point of inflection**, called the **critical point**. At this temperature, the distinction between liquid and gas vanishes, and CO_2 exists as a **critical fluid**. Beyond this temperature, no amount of pressure can liquefy the gas.

(Experiment with suitable explanation -7 marks, diagram 3 marks)

18. Discuss the Lowry-Bronsted theory and the Lewis theory of acids and bases

The Lowry–Brønsted theory defines an acid as a proton (H^+) donor and a base as a proton (H^+) acceptorAccording to this theory, every acid–base reaction involves the transfer of a proton from the acid to the base. For example, in the reaction $HCl + H_2O \rightarrow H_3O^+ + Cl^-$, HCl donates a proton and acts as an acid, while water accepts the proton and acts as a base. This theory also introduces the concept of conjugate acid–base pairs, where an acid forms its conjugate base after donating a proton, and a base forms its conjugate acid after accepting a proton (e.g., HCl/Cl^- and H_2O/H_3O^+). However, the Lowry–Brønsted concept is limited to reactions involving proton transfer and cannot explain acid–base reactions where no proton is involved. **(Definition – 2 marks, Explanation with example – 3 marks)**

The Lewis theory provides a more general definition, where an acid is a species that accepts an electron pair and a base is a species that donates an electron pair. (Definition – 2 marks) For example, in the reaction $BF_3 + NH_3 \rightarrow BF_3 \leftarrow NH_3$, boron trifluoride ($BF_3$) acts as a Lewis acid by accepting an electron pair from ammonia (NH_3), which acts as a Lewis base, forming a coordinate covalent bond. The Lewis theory is more comprehensive because it includes reactions that do not involve protons and explains a wider range of chemical behavior, including complex formation and catalytic reactions. **(Definition – 2 marks, Explanation with example – 3 marks)**

19. **Explain a simple eutectic system using lead-silver as an example. Apply phase rule.**

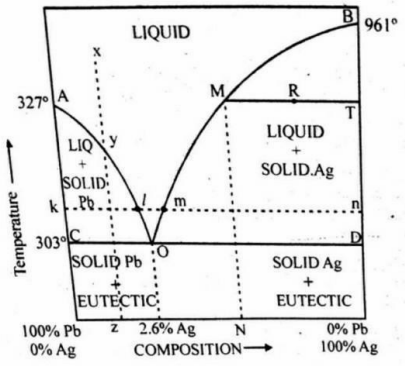
The **lead–silver (Pb–Ag) system** is a **simple eutectic system** consisting of two components that are completely miscible in the liquid state but almost immiscible in the solid state. The diagram plots **temperature on the y-axis** and **composition on the x-axis** (from 100% Pb to 100% Ag).At the top region **A–B**, the system exists entirely in the **liquid state**. The two curves **A–O** and **B–O** are called **liquidus curves**, which represent the temperatures at which solidification begins. As a molten mixture cools, the behavior depends on its composition.For a **lead-rich alloy (left side)**, cooling from point A follows the liquidus curve A–O. At point **y**, solid lead (Pb) starts to crystallize, and the system enters a two-phase region **(liquid + solid Pb)**. As cooling continues, more Pb separates out and the liquid becomes richer in silver until it reaches point **O**, the **eutectic point**.For a **silver-rich alloy (right side)**, cooling follows curve B–O. At point **R**, solid silver (Ag) begins to separate, and the system enters the region **(liquid + solid Ag)**. With further cooling, more silver crystallizes, and the liquid composition shifts toward the eutectic composition until point O is reached.The point **O (2.6% Ag, 303°C)** is the **eutectic point**, where the liquid transforms simultaneously into a fine mixture of **solid Pb and solid Ag**. This occurs at a constant temperature, and below this temperature, the system consists entirely of solid phases.The

horizontal line **C–D (303°C)** represents the **eutectic temperature**, below which only solid phases exist. The regions below this line are:

- Left side: **Solid Pb + eutectic mixture**
- Right side: **Solid Ag + eutectic mixture**

Lead-silver system and its explanation: Phase Diagram

Lead-silver system is a two component system. The phase diagram of this system is shown in fig. Lead and silver are immiscible with each other in liquid state. And they do not react with each other.



Thus, the diagram clearly shows three important regions:

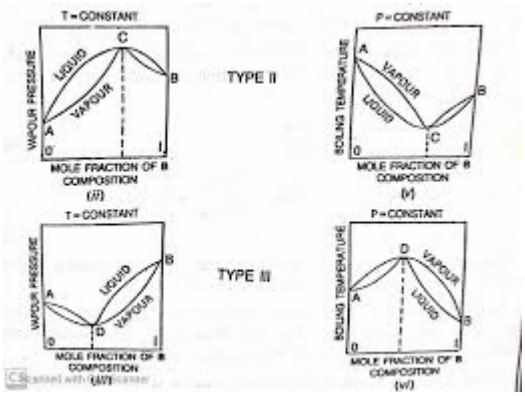
1. **Liquid region (above liquidus curves)**
2. **Two-phase regions (liquid + solid Pb or liquid + solid Ag)**
3. **Solid region (below eutectic line)**

Applying the **Gibbs phase rule** for a condensed system: F = C − P + 1. Here, number of components C = 2 (Pb and Ag). At the eutectic point, three phases coexist (liquid + solid Pb + solid Ag), so P = 3. Therefore,F = 2 − 3 + 1 = 0, indicating that the system is **invariant**, meaning temperature and composition are fixed.

This system is important in metallurgy, especially in the **Pattinson process**, where silver is separated from lead by controlled cooling and crystallization based on this phase behavior.***(Diagram -2.5 marks, Explanation – 5 marks, Phase rule application – 2.5 marks)***

20. **Evaluate a Minimum Boiling Azeotrope and a Maximum Boiling Azeotrope by relating them to their respective types of Raoult's Law deviation.**

A **Type II non-ideal solution** shows Large **positive deviation from Raoult's law**, meaning the total vapour pressure of the solution is **greater than predicted** for an ideal solution. This occurs because the **intermolecular forces between unlike molecules (A–B)** are **weaker** than those between like molecules (A–A and B–B), so molecules escape more easily into the vapour phase.In the **vapour pressure–composition curve**, the experimental curve lies **above the Raoult's law straight line**, indicating higher vapour pressure at all compositions.In the **temperature–composition (boiling point) curve**, the solution shows a **minimum boiling point**, because higher vapour pressure leads to lower boiling temperature.**Examples:** Ethanol–water, acetone–carbon disulfide.A **Type III non-ideal solution** shows large **negative deviation from Raoult's law**, where the observed vapour pressure is **lower than predicted**. This is due to **stronger intermolecular forces between unlike molecules (A–B)** compared to like molecules, reducing the tendency of molecules to escape into vapour phase. In the **vapour pressure–composition curve**, the actual curve lies **below the Raoult's law line**, indicating reduced vapour pressure. In the **temperature–composition curve**, this leads to a **maximum**



boiling point, since lower vapour pressure requires higher temperature for boiling. **Example:** Acetone–chloroform system.

Minimum Boiling Azeotrope and a Maximum Boiling Azeotrope is out of syllabus, so Reasonable attempt of type II and type III non ideal solutions with suitable explanation and curves can be give full credit.